TRUSTED® COMPUTING GROUP

SPECIFICATION

TCG Storage Opal Family Feature Set: Logical Port Locking

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1 Scope

This specification defines the Logical Port Locking Feature Set.

1.1 Key Words

The key words “MUST,” “MUST NOT,” “REQUIRED,” “SHALL,” “SHALL NOT,” “SHOULD,” “SHOULD NOT,” “RECOMMENDED,” “MAY,” and “OPTIONAL” in this document normative statements are to be interpreted as described in RFC-2119, Key words for use in RFCs to Indicate Requirement Levels (see [1]).

1.2 Statement Type

Please note a very important distinction between different sections of text throughout this document. There are two distinctive kinds of text: informative comment and normative statements. Because most of the text in this specification will be of the kind normative statements, the authors have informally defined it as the default and, as such, have specifically called out text of the kind informative comment. They have done this by flagging the beginning and end of each informative comment and highlighting its text in gray. This means that unless text is specifically marked as of the kind informative comment, it can be considered a kind of normative statement.

Start of informative comment

This is the first paragraph of 1–n paragraphs containing text of the kind *informative comment* ...

This is the second paragraph of text of the kind *informative comment* ...

This is the nth paragraph of text of the kind *informative comment* ...

To understand the TCG specification the user must read the specification. (This use of MUST does not require any action).

End of informative comment

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2 Introduction and Concepts

2.1 Document Purpose

The Storage Workgroup specifications provide a comprehensive architecture for Storage Devices (SDs) under policy control as determined by the trusted platform host, the capabilities of the Storage Devices to conform to the policies of the trusted platform, and the lifecycle state of the SD as a Trusted Peripheral.

2.2 Intended Audience

The intended audience for this specification is both trusted Storage Device manufacturers and developers that want to use these Storage Devices in their systems.

2.3 Conventions

2.3.1 Font Conventions

Names of methods and Security Provider (SP) tables are in `Courier New font` (e.g., the `Set` method, the `Locking` table). This convention does not apply to method and table names appearing in headings or captions.

Hexadecimal numbers are in `Courier New font`. All other text is in the Arial font.

2.3.2 Security Provider (SP) Table Cell Symbols Legend

The legend in Table 1 defines the SP table cell symbols used to communicate Read-Write and Access Control attributes of a cell. These symbols are informative only. Unless otherwise indicated, the table cell content is normative. If a table cell is empty or blank (regardless of symbols), then the contents of that cell are not specified in this specification. The contents might be specified in another specification.

Start of Informative Comment

As specified in the Core Specification (see [2] section 3.4.2 and section 5.3.4.3), Access Control limits the methods that can be processed on an SP, a table, or on specific rows and columns of a table. Access Control is specified in layers. The top layer of the mechanism is an Access Control List (ACL). ACLs contain lists of Access Control Elements (ACEs). ACEs contain Boolean combinations of authorities. ACLs and ACEs bind methods to the authorities that are permitted to invoke them. When the state of authorities specified in an ACE Boolean expression are satisfied as described in the Core Specification (see [2] section 3.4.2.1) (for example, the ACE Boolean expression resolves to True on successful authentication), the ACE permits methods to be invoked on an SP, a table, or on specific rows and columns of a table based on the ACL that contains the ACE.

End of Informative Comment

The “R-W” column in Table 1 describes if the contents of an SP table cell is read-only, writeable, or not required (NR). This cell property can be determined through the mapping of methods, that provide read or write access to the cell, and an ACL in the `AccessControl` table (see Core Specification [2] section 5.3.4.3). If a cell is readable, then there exists a method and associated ACE definition to retrieve the contents of the cell. If a cell is writeable, then there exists a method and associated ACE definition to modify the contents of the cell. There MAY be other conditions that prevent a method from being invoked on a table cell outside the scope of the Access Control specified in the `AccessControl` table.

The “Value” column in Table 1 describes if the contents of an SP table cell is specified by the feature set, vendor unique (VU), or not required (NR). When a cell content is VU, the cell contents contain the text “VU” and access control for the cell is defined by the feature set. Access Control for a cell is not defined (ND) if no ACE defined by this specification is associated with the cell. If the cell contents are (NR), the `Get` method MAY omit the column from a method response.

The “Access Control” column in [2] describes if the access control associated with an SP table cell is user-personalizable, fixed, or not defined (ND). Access Control for a cell is personalizable if the Boolean expression of the

ACEs associated with the cell are writable. Access Control for a cell is fixed if the Boolean expression of the ACEs associated with the cell are not writeable. Access Control for a cell is not defined (ND) if there does not exist an ACE that is associated with the cell. If the cell contents are (NR), the Get method MAY omit the column from a method response. The behavior of AddACE and RemoveACE methods with respect to ACLs is not defined.

Table 1 - SP Table Legend

Table Cell Symbols Legend	R-W	Value	Access Control	Comment
Read-only Fixed Access Control (RF)	Read-only	Logical Port Locking Feature Set specified	Fixed	• Cell content is Read-Only. • Access control is fixed. • Value is specified by the Configurable Locking for NVMe Namespaces and SCSI LUNs Feature Set
Read-only Fixed Access Control-VU (RFV)	Read-only	VU	Fixed	• Cell content is Read-Only. • Access Control is fixed. • Values are Vendor Unique (VU). A minimum or maximum value might be specified.
Undefined (UD)	Not Defined (ND)	Not Required (NR)	Not Defined (ND)	• Cell content is (NR). • Access control is not defined (ND). • Any text in table cell is informative only. • A `Get` MAY omit this column from the method response.
Writable Personalizable Access Control (WP)	Write	Preconfigured/ user personalizable, or VU	Preconfigured, user personalizable	• Cell content is writable. • Access control is personalizable • `Get` Access Control is not described by this symbol • Cell contents preconfiguration for VU column is “VU”.

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Table Cell Symbols Legend	R-W	Value	Access Control	Comment
Writable Fixed Access Control (WF)	Write	Preconfigured/ user personalizable, or VU	Fixed	• Cell content is writable. • Access control is fixed. • `Get` Access Control is not described by this symbol • Cell contents preconfiguration for VU column is “VU”.

2.3.3 List Conventions

2.3.3.1 Lists Overview

Lists are associated with an introductory paragraph or phrase and are numbered relative to that paragraph or phrase (i.e., all lists begin with an a) or 1) entry).

Each item in a list is preceded by an identification with the style of the identification being determined by whether the list is intended to be an ordered list or an unordered list.

If the item in a list is not a complete sentence, the first word in the item is not capitalized. If the item in a list is a complete sentence, the first word in the item is capitalized.

Each item in a list ends with a semicolon, except the last item, which ends in a period. The next to the last entry in the list ends with a semicolon followed by an “and” or an “or” (i.e., “…; and”, or “…; or”). The “and” is used if all the items in the list are required. The “or” is used if only one or more items in the list are required.

2.3.3.2 Unordered Lists

An unordered list is one in which the order of the listed items is unimportant (i.e., it does not matter where in the list an item occurs as all items have equal importance). In an unordered list, each item starts with a lowercase letter followed by a close parenthesis. If it is necessary to subdivide a list item further with an additional unordered list (i.e., have a nested unordered list), then the nested unordered list is indented and each item in the nested unordered list starts with an uppercase letter followed by a close parenthesis.

The following is an example of an unordered list with a nested unordered list:

EXAMPLE - The following are the items for the assembly:
a) a box containing:
    A) a bolt;
    B) a nut; and
    C) a washer;
b) a screwdriver; and
c) a wrench.

2.3.3.3 Ordered Lists

An ordered list is one in which the order of the listed items is important (i.e., item n is required before item n+1). In an ordered list, each item starts with a Western-Arabic numeral followed by a close parenthesis. If it is necessary to subdivide a list item further with an additional unordered list (i.e., have a nested unordered list), then the nested

unordered list is indented and each item in the nested unordered list starts with an uppercase letter followed by a close parenthesis.

The following is an example of an ordered list with a nested unordered list:

EXAMPLE - The following are the instructions for the assembly:
1) remove the contents from the box;
2) assemble the item;
   A) use a screwdriver to tighten the screws; and
   B) use a wrench to tighten the bolts;
   and
3) take a break.

2.3.4 Numbering Conventions

A binary number is represented in this standard by any sequence of digits consisting of only the Western-Arabic numerals 0 and 1 immediately followed by a lowercase b (e.g., 0101b). Underscores or spaces might be included between characters in binary number representations to increase readability or delineate field boundaries (e.g., 0 0101 1010b or 0_0101_1010b).

A hexadecimal number is represented in this standard by any sequence of digits consisting of only the Western-Arabic numerals 0 through 9 and/or the uppercase English letters A through F immediately preceded by "`0x`". Underscores or spaces might be included between characters in hexadecimal number representations to increase readability or delineate field boundaries (e.g., `0xFD8C FA23` or `0x0B_FD8C_FA23`). Hexadecimal numbers are in `Courier New font`.

A decimal number is represented in this standard by any sequence of digits consisting of only the Western-Arabic numerals 0 through 9 not immediately followed by a lowercase b or lowercase h (e.g., 25). This standard uses the following conventions for representing decimal numbers:

a) the decimal separator (i.e., separating the integer and fractional portions of the number) is a period;
b) the thousands separator (i.e., separating groups of three digits in a portion of the number) is a space; and
c) the thousands separator is used in both the integer portion and the fraction portion of a number.

A decimal number represented in this standard with an overline over one or more digits following the decimal point is a number where the overlined digits are infinitely repeating (e.g., $666.\overline{6}$ means 666.666 666… or 666 2/3, and $12.\overline{142857}$ means 12.142 857 142 857… or 12 1/7).

2.3.5 Bit Conventions

Name (n:m), where n is greater than m, denotes a set of bits (e.g., Feature (7:0)).

2.3.6 Number Range Conventions

p..q, where p is less than q, represents a range of numbers (e.g., words 100..103 represents words 100, 101, 102, and 103).

2.3.7 Specify and Indicate Conventions

In a given message, command, or other information exchange between a requestor (e.g., a host) and a responder (e.g., an SD):

a) the requestor specifies information in the request; and
b) the responder indicates information in the response.

2.4 Document References

2.4.1 Document Precedence

If there is a conflict between this specification and any other reference, then the precedence is (where a lower number indicates higher precedence):

1. this specification; and
2. normative references (see section 2.4.2).

2.4.2 Normative References

[1] IETF RFC 2119, 1997, “Key words for use in RFCs to Indicate Requirement Levels”

[2] Trusted Computing Group (TCG), “TCG Storage Architecture Core Specification”, Version 2.01

[3] Trusted Computing Group (TCG), “Storage Interface Interactions Specification”, Version 1.11

[4] Trusted Computing Group (TCG), “TCG Storage Security Subsystem Class: Opal”, Version 2.02

[5] Trusted Computing Group (TCG), “TCG Storage Security Subsystem Class: Opalite”, Version 1.00

[6] Trusted Computing Group (TCG), “TCG Storage Security Subsystem Class: Ruby”, Version 1.00

[7] Trusted Computing Group (TCG), “TCG Storage Security Subsystem Class: Pyrite”, Version 2.01

[8] National Institute of Standards and Technology, FIPS PUB 140-3 SECURITY REQUIREMENTS FOR CRYPTOGRAPHIC MODULES

[9] Trusted Computing Group (TCG), “Storage Interface Interactions Specification”, Version 1.20

2.4.3 Informative References

This feature set does not refer to any informative document.

2.5 Dependencies on Other Feature Sets

This feature set does not depend upon any other feature sets.

2.6 Interactions with Other Feature Sets

This feature set has no interactions with other feature sets.

2.7 Definition of Terms

Term	Definition
FW	Firmware
IF-RECV	An interface command used to retrieve security protocol data from the TPer (see [3])
IF-SEND	An interface command used to transmit security protocol data to the TPer (see [3])
IF-FWCOMMIT	An interface command used to update FW on the Storage Device (see [9])
IF-DIAG	An interface command used to send manufacturer diagnostic commands to the Storage Device. Which commands are classified as IF-DIAG commands is vendor unique and is outside the scope of this specification.

Term	Definition
IF-LOG	An interface command used to retrieve manufacturer specific logs from the Storage Device. Which commands are classified as IF-LOG commands is vendor unique and is outside the scope of this specification.
Issuance	The act of creating an SP on a TPer from one or more templates (see [2]).
Admin SP	A security provider that incorporates the Admin Template as described in the Core Spec (see [2])
Locking SP	A security provider that incorporates the Locking Template as described in the Core Spec (see [2])
ND	Not Defined.
NR	Not Required.
Opal family	Any of: Opal SSC, Opalite SSC, Ruby SSC, or Pyrite SSC
Rollback FW Update	A FW update where the incoming FW, if accepted by the Storage Device, would rollback the FW to a lower security revision. How a Storage Device manufacturer determines or assigns security revision numbers to specific FW is outside the scope of this specification. Enabling rollback to a lower security revision presents a security risk. Some device manufacturers might implement additional behaviors to mitigate this risk (e.g., a successful Rollback FW Update would perform a cryptographic erase of User Data).
SD	Storage Device
SP	Security Provider
SSC	Security Subsystem Class. SSC specifications describe profiled sets of TCG functionality
TPer	The Trusted Peripheral is the subset of a Storage Device for which TCG manages security
VU	Vendor Unique, these values are unique to each SD manufacturer

3 Logical Port Locking Feature Set Overview

This Logical Port Locking Feature Set provides mechanisms to enforce access controls on specific services on a Storage Device using the concept of Logical Ports. If a Logical Port is unlocked, the associated service is enabled or available on the Storage Device and if the Logical Port is locked, the associated service is disabled or unavailable. The Logical Ports defined by this specification are described in section 5.1.

Providing the ability to enable access control for device services enables TCG devices to comply with other security requirements or recommendations such as [8].

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4 SSC Specific Functionality

This section specifies the additional SSC-specific functionality (not contained in the Core Specification [2] or the applicable Opal Family SSC) required to support this feature set.

4.1 Methods

This section defines the new methods and modifications to existing methods required for this feature set.

4.1.1 New Methods

There are no new methods defined by this feature set.

4.1.2 Modified Methods

If the TPer supports the ACE_SID_PSID_Set_Port ACE object or the ACE_SID_Makers_Set_Port ACE object (see section 5.3.1.3), then the value of the MaxAuthentications field of the `Properties` method response SHALL be greater than or equal to 3.

Start of Informative Comment

The Opal Family SSCs require MaxAuthentications field of the `Properties` method response to be greater than or equal to 2. To properly support the ACE_SID_PSID_Set_Port ACE object and ACE_SID_Makers_Set_Port ACE object, the MaxAuthentications field of the `Properties` method response needs to be greater than or equal to 3 so that the Anybody authority (always authenticated) and two additional authorities can be authenticated at the same time.

End of Informative Comment

4.2 Tables

This section defines the new tables and modifications to existing tables required for this feature set.

4.2.1 New Tables

This section defines the new tables required for this feature set.

4.2.1.1 Logical_Port (Object Table)

The TPer SHALL support the `Logical_Port` table as defined in Table 2. The UID for the `Logical_Port` table SHALL be 0x00 00 08 08 00 00 00 00.

Table 2 – Logical_Port Table

Column Number	Column Name	IsUnique	Column Type
0x00	UID	No	uid
0x01	Name	Yes	name
0x02	CommonName	Yes	name
0x03	LockOnReset	No	reset_types
0x04	PortLocked	No	boolean

4.2.1.1.1 UID

The value of this column is the unique identifier of this row in the `Logical_Port` table.

This column SHALL NOT be modifiable by the host.

4.2.1.1.2 Name

The value of this column is the name of the Logical_Port object.

For Logical_Port objects that exist at issuance, this column SHALL NOT be modifiable by the host.

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4.2.1.1.3 CommonName
This is a name that MAY be shared by multiple Logical_Port objects.

For Logical_Port objects that exist at issuance, this column SHALL NOT be modifiable by the host.

4.2.1.1.4 LockOnReset
The value of this column defines the locking behavior of this Logical_Port object at reset, dependent on reset type. The values enumerated in this column SHALL identify the reset types that cause the values of the PortLocked column of the Logical_Port object to be set to True.

An empty list (“[ ]”) SHALL indicate that PortLocked column does not change on any reset.

The TPer SHALL support the following LockOnReset column values:

a) [ ] (i.e., the empty list); and
b) [0] (i.e., Power Cycle).

The TPer MAY support other values for the LockOnReset column.

4.2.1.1.5 PortLocked
The value of this column identifies the current lock state of the Logical_Port object. The TPer behavior with respect to the value of this column is dependent on the specific Logical_Port object (see section 5.3.1.2).

4.2.2 Modified Tables
This feature set modifies the following tables:

- Admin SP `Table` table
- Admin SP `ACE` table
- Admin SP `AccessControl` table

See section 5.3.1 for details.

4.3 Types
This section defines new types and modifications to existing types required for this feature set.

4.3.1 New Types
There are no new types defined by this feature set.

4.3.2 Modified Types
If the TPer supports the ACE_SID_PSID_Set_Port ACE object (see section 5.3.1.3), then the boolean_ACE type SHALL support the AND Boolean operator for the value of the BooleanExpr column of the ACE_SID_PSID_Set_Port ACE object .

If the TPer supports the ACE_SID_Makers_Set_Port ACE object (see section 5.3.1.3), then the boolean_ACE type SHALL support the AND Boolean operator for the value of the BooleanExpr column of the ACE_SID_Makers_Set_Port ACE object.

Start of Informative Comment

The BooleanExpr column of the ACE_SID_PSID_Set_Port ACE object and the ACE_SID_Makers_Set_Port ACE object contains the AND operator. Support for the Boolean AND operator is not required by the Opal Family SSCs.

End of Informative Comment

5 Feature Set Requirements

This section defines the requirements for this feature set.

5.1 Requirements Overview

If the Storage Device supports this feature set, the Storage Device SHALL support at least one of the following Logical Ports:

- Diagnostic: This port SHALL be used to provide access control for manufacturer diagnostic commands on the Storage Device via IF-DIAG commands (see section 5.1.1).
- FWDownload: This port SHALL be used to provide access control for FW updates of the Storage Device via IF-FWCOMMIT commands (see section 5.1.2). This port SHALL be supported if the RBFWDownload port is supported.
- RBFWDownload: This port SHALL be used to provide access control for Rollback FW Updates of the Storage Device via IF-FWCOMMIT commands (see section 5.1.2).
- ManufacturerLog: This port SHALL be used to provide access control for manufacturer specific logs on the Storage Device via IF-LOG commands (see section 5.1.3).

The Logical Ports are represented in the TCG data structures via the Logical_Port objects (see section 4.2.1.1).

5.1.1 Interactions with IF-DIAG Commands

Start of Informative Comment

The following rule specifies the successful operation of an IF-DIAG command when the Diagnostic Logical_Port object is unlocked.

End of Informative Comment

If:

- the Storage Device receives an IF-DIAG command;
- the IF-DIAG command would otherwise succeed;
- the Storage Device supports the Diagnostic Logical_Port object; and
- the value of the PortLocked column of the Diagnostic Logical_Port object is False,

then, the Storage Device SHALL process the IF-DIAG command.

Start of Informative Comment

The following rule prevents the Storage Device from accepting a manufacturer diagnostic command if the Diagnostic Logical_Port object is locked.

End of Informative Comment

If:

- the Storage Device receives an IF-DIAG command;
- the Storage Device supports the Diagnostic Logical_Port object; and
- the value of the PortLocked column of the Diagnostic Logical_Port object is True,

then, the Storage Device SHALL reject the IF-DIAG command with an interface status code of Data Protection Error (see [3]).

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5.1.2 Interactions with IF-FWCOMMIT Commands

Start of Informative Comment

The following rules specify the successful operation of an IF-FWCOMMIT command when the FWDownload Logical_Port object is unlocked and the RBFWDownload Logical_Port is not supported.

End of Informative Comment

If:

- the Storage Device receives an IF-FWCOMMIT command;
- the Storage Device has received a complete FW download image;
- the IF-FWCOMMIT command would otherwise succeed;
- the Storage Device supports the FWDownload Logical_Port object;
- the Storage Device does not support the RBFWDownload Logical_Port object; and
- the value of the PortLocked column of the FWDownload Logical_Port object is False,

then, the Storage Device SHALL process the IF-FWCOMMIT command.

Start of Informative Comment

The following rules specify the successful operation of an IF-FWCOMMIT command when the FWDownload Logical_Port object is unlocked, the RBFWDownload Logical_Port is supported, and the FW update is not a Rollback FW Update.

End of Informative Comment

If:

- the Storage Device receives an IF-FWCOMMIT command;
- the Storage Device has received a complete FW download image;
- the IF-FWCOMMIT command would otherwise succeed;
- the Storage Device supports the FWDownload Logical_Port object;
- the Storage Device supports the RBFWDownload Logical_Port object;
- the value of the PortLocked column of the FWDownload Logical_Port object is False; and
- the FW update is not a Rollback FW Update,

then, the Storage Device SHALL process the IF-FWCOMMIT command.

Start of Informative Comment

The following rule prevents the updating of the Storage Device FW if the FWDownload Logical_Port object is locked.

End of Informative Comment

If:

- the Storage Device receives an IF-FWCOMMIT command;
- the Storage Device has received a complete FW download image;
- the Storage Device supports the FWDownload Logical_Port object; and
- the value of the PortLocked column of the FWDownload Logical_Port object is True,

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then, the Storage Device SHALL reject the IF-FWCOMMIT command with an interface status code of Data Protection Error (see [3]).

Start of Informative Comment

The following rule specifies the successful operation of an IF-FWCOMMIT command when the FWDownload Logical_Port object is unlocked and the FW update is a Rollback FW Update.

End of Informative Comment

If:

- the Storage Device receives an IF-FWCOMMIT command;
- the Storage Device has received a complete FW download image;
- the IF-FWCOMMIT command would otherwise succeed;
- the Storage Device supports the FWDownload Logical_Port object;
- the Storage Device supports the RBFWDownload Logical_Port object;
- the value of the PortLocked column of the FWDownload Logical_Port object is False;
- the value of the PortLocked column of the RBFWDownload Logical_Port object is False; and
- the FW update is a Rollback FW Update,

then, the Storage Device SHALL process the IF-FWCOMMIT command.

Start of Informative Comment

The following rule prevents the updating of the Storage Device FW to FW with a lower security revision if the RBFWDownload Logical_Port object is locked.

End of Informative Comment

If:

- the Storage Device receives an IF-FWCOMMIT command;
- the Storage Device has received a complete FW download image;
- the Storage Device supports the FWDownload Logical_Port object;
- the Storage Device supports the RBFWDownload Logical_Port object;
- the value of the PortLocked column of the FWDownload Logical_Port object is False;
- the value of the PortLocked column of the RBFWDownload Logical_Port object is True; and
- the FW update is a Rollback FW Update,

then, the Storage Device SHALL reject the IF-FWCOMMIT command with an interface status code of Data Protection Error (see [3]).

5.1.3 Interactions with IF-LOG Commands

Start of Informative Comment

The following rule specifies the successful operation of an IF-LOG command when the ManufacturerLog Logical_Port object is unlocked.

End of Informative Comment

If:

- the Storage Device receives an IF-LOG command;
- the IF-LOG command would otherwise succeed;
- the Storage Device supports the ManufacturerLog Logical_Port object; and
- the value of the PortLocked column of the ManufacturerLog Logical_Port object is False,

then, the Storage Device SHALL process the IF-LOG command.

Start of Informative Comment

The following rule prevents the Storage Device FW from accepting a manufacturer specific log command if the ManufacturerLog Logical_Port object is locked.

End of Informative Comment

If:

- the Storage Device receives an IF-LOG command;
- the Storage Device supports the ManufacturerLog Logical_Port object; and
- the value of the PortLocked column of the ManufacturerLog Logical_Port object is True,

then, the Storage Device SHALL reject the IF-LOG command with an interface status code of Data Protection Error (see [3]).

5.2 Level 0 Discovery

A Storage Device implementing the Logical Port Locking Feature Set SHALL:

a) return the Logical Port Locking Feature Descriptor as defined in section 5.2.1; and

b) support the Level 0 Discovery response requirements defined in the applicable Opal Family SSC.

5.2.1 Logical Port Locking Feature Descriptor (Feature Code = 0x040B)

The Logical Port Locking Feature Descriptor SHALL be returned when the Storage Device supports the Logical Port Locking Feature Set. The contents of the feature descriptor are defined in Table 3.

Table 3 - Level 0 Discovery – Logical Port Locking Feature Descriptor

Byte \ Bit	7	6	5	4	3	2	1	0
0	(MSB)			Feature Code				
1								(LSB)
2	Feature Descriptor Version Number				Feature Set Minor Version Number			
3	Length							
4				Reserved				
…								
7								
8	(MSB)			Number of Logical_Port Objects				
…								
11								(LSB)
12				Reserved				
…								
15								

5.2.1.1 Feature Code
The value of the Feature Code field value SHALL be set to 0x040B.

5.2.1.2 Feature Descriptor Version Number
The value of the Feature Descriptor Version Number field SHALL be set to 0x1.

5.2.1.3 Feature Set Minor Version Number
The value of the Feature Set Minor Version Number represents the minor version of the Logical Port Locking Feature Set supported by the Storage Device.

The value of the Feature Set Minor Version Number field SHALL be set as specified in Table 4.

Table 4 - Feature Set Minor Versions

Feature Set Minor Version Number	Specification Referenced
0x0	Logical Port Locking Feature Set v1.00
All others	Reserved

5.2.1.4 Length
The value of the Length field indicates the number of bytes in the descriptor(s) following byte 3. The value of the Length field SHALL be 12.

5.2.1.5 Number of Logical_Port Objects
The value of the Number of Logical Port Objects field indicates the total number of Logical_Port objects supported in the Logical Port table (see section 5.3.1.2). The minimum value of the Number of Logical_Port Objects field SHALL be 1.

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5.3 Admin SP

In addition to the Admin SP requirements defined in the Core Specification [2] or the applicable Opal Family SSC, a Storage Device implementing this feature set SHALL support the additions to the Admin SP specified in this section.

5.3.1 Tables

5.3.1.1 Table Table

The TPer SHALL support additional rows on the `Table` Table and preconfigure those additional rows as defined in Table 5.

The conventions used by Table 5 are defined in section 2.3.2.

Table 5 - Admin SP – Table Table Preconfiguration

UID[RF]	Name[RF]	CommonName	TemplateID	CommonName	Kind[RF]	Column	NumColumns	Rows	RemoveACEACL	RowsFree	RowBytes	LastID	MinSize	MaxSize
00 00 00 01 00 00 08 08	Logical_Port[1]	(NR)[2]	(NR)[2]	(NR)[2]	Object[1]	(NR)[2]	(NR)[2]	(NR)[2]	(NR)[2]	(NR)[2]	(NR)[2]	(NR)[2]	(NR)[2]	(NR)[2]

Notes:
1. The text in the table cell is informative. See the Core Specification [2] and the applicable Opal Family SSC.
2. The content of the cell is defined by the Core Specification [2] and the applicable Opal Family SSC.
3. RF: The content of each cell in the column is Read-only as specified by section 5.3.1.4.

5.3.1.2 Logical Port Table

For each Logical_Port object supported by the TPer (see section 5.1), the TPer SHALL support the associated rows on the `Logical_Port` table (see section 4.2.1.1) and preconfigure those additional rows as defined in Table 6.

The conventions used by Table 6 are defined in section 2.3.2.

Table 6 - Admin SP – Logical_Port Table Preconfiguration

UID[1, RF]	Name[2]	CommonName[3]	LockOnReset[WP]	PortLocked[WP]
00 00 08 08 00 00 00 01	Diagnostic	(NR)	[0]	True
00 00 08 08 00 00 00 02	FWDownload	(NR)	VU	VU
00 00 08 08 00 00 00 03	RBFWDownload	(NR)	VU	VU
00 00 08 08 00 00 00 04	ManufacturerLog	(NR)	VU	VU

Notes:
1. Support of each row in this column is optional.
2. The text in each cell of this column is informative. See the Core Specification [2] and the applicable Opal Family SSC.
3. The content of each cell in this column is defined by the Core Specification [2] and the applicable Opal Family SSC.
4. RF: The content of each cell in this column is Read-only as specified by section 5.3.1.4.
5. WP: The content of each cell in this column is writeable using the `Set` method.

5.3.1.3 ACE Table

The TPer SHALL support additional rows on the `ACE` table and preconfigure those additional rows as defined in Table 7.

The conventions used by Table 7 are defined in section 2.3.2.

Table 7 - Admin SP – ACE Table Preconfiguration

Table Association – Informative only	UID[3, RF]	Name	CommonName[2]	BooleanExpr[RF]	Columns[RF]
Logical_ Port[1]					
	00 00 00 08 00 06 30 01	“ACE_SID_Makers_Set_Port”	(NR)	SID AND Makers	LockOnReset, PortLocked

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Table Association – Informative only	UID[3, RF]	Name	CommonName[2]	BooleanExpr[RF]	Columns[RF]
	00 00 00 08 00 06 30 02	“ACE_SID_Set_Port”	(NR)	SID	LockOnReset, PortLocked
	00 00 00 08 00 06 30 03	“ACE_SID_PSID_Set_Port”	(NR)	SID AND PSID	LockOnReset, PortLocked
	00 00 00 08 00 06 30 04	“ACE_PSID_Set_Port”	(NR)	PSID	LockOnReset, PortLocked

Notes:
1. The text in the table cell is informative. See the Core Specification [2] and the applicable Opal Family SSC.
2. The content in each cell of this column is defined by the Core Specification [2] and the applicable Opal Family SSC.
3. Support of each row in this column is only required if the ACE is required in the `AccessControl` table (see section 5.3.1.4).
4. RF: The content of each cell in the column is Read-only as specified by section 5.3.1.4.

5.3.1.4 AccessControl Table

The TPer SHALL support additional rows in the `AccessControl` table and preconfigure those additional rows as defined in Table 8.

The conventions used by Table 8 are defined in section 2.3.2.

Start of Informative Comment

The following describes the rationale for the preconfigured settings for Logical_Port objects in the `AccessControl` table.

Diagnostic: The use of device diagnostics commands could affect the security of the TPer such as exposing sensitive data or changing the device configuration. Only the device owner (i.e., SID) or the device manufacturer with permission from the device owner ought to be able to access the device diagnostic commands (i.e., Makers and SID).

FWDownload: Modifying the device FW ought to only occur with permission of the device owner (i.e., SID).

RBFWDownload: Modifying the device FW such that the new FW will have a lower security revision than the resident FW ought to only occur with permission of the device owner (i.e., SID), physical presence (i.e., PSID), or device owner and physical presence (i.e., SID and PSID).

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ManufacturerLog: The device logs might contain sensitive data so the logs ought to only be accessible with device owner permission (i.e., SID).

If there are options for `Access Control` table pre-configuration with respect to Logical_Port objects, what a device implements is determined by the device manufacturer.

End of Informative Comment

Table 8 - Admin SP – AccessControl Table Preconfiguration

Table Association – informative only	**UID**	**InvokingID**[RF]	InvokingID Name	**MethodID**	**CommonName**	**ACL**[RF]	**Log**	**AddACEACL**	**RemoveACEACL**	**GetACLACL**[RF]	**DeleteMethodACL**	**AddACELog**	**RemoveACELog**	**GetACLLog**	**DeleteMethodLog**	**LogTo**
ACE[1]																
	(ND)[3]	00 00 00 08 00 06 30 01[4]	ACE_SID_Makers_Set_Port 1	Get	(NR)[2]	ACE_Anybody	(NR)[2]	(NR)[2]	(NR)[2]	ACE_Anybody	(NR)[2]	(NR)[2]	(NR)[2]	(NR)[2]	(NR)[2]	(NR)[2]
	(ND)[3]	00 00 00 08 00 06 30 02[4]	ACE_SID_Set_Port[1]	Get	(NR)[2]	ACE_Anybody	(NR)[2]	(NR)[2]	(NR)[2]	ACE_Anybody	(NR)[2]	(NR)[2]	(NR)[2]	(NR)[2]	(NR)[2]	(NR)[2]
	(ND)[3]	00 00 00 08 00 06 30 03[4]	ACE_SID_PSID_Set_Port[1]	Get	(NR)[2]	ACE_Anybody	(NR)[2]	(NR)[2]	(NR)[2]	ACE_Anybody	(NR)[2]	(NR)[2]	(NR)[2]	(NR)[2]	(NR)[2]	(NR)[2]
Logical_Port[1]																

			Table Association – informative only
(ND)[3]	(ND)[3]	(ND)[3]	**UID**
00 00 08 08 00 00 00 01[5]	00 00 08 08 00 00 00 01[5]	00 00 08 08 00 00 00 00	**InvokingID**[RF]
Diagnostic[1]	Diagnostic[1]	Logical_Port_Locking[1]	InvokingID Name
Get	Set	Next	**MethodID**
(NR)[2]	(NR)[2]	(NR)[2]	**CommonName**
ACE_Anybody	Note 10	ACE_Anybody	**ACL**[RF]
(NR)[2]	(NR)[2]	(NR)[2]	**Log**
(NR)[2]	(NR)[2]	(NR)[2]	**AddACEACL**
(NR)[2]	(NR)[2]	(NR)[2]	**RemoveACEACL**
ACE_Anybody	ACE_Anybody	ACE_Anybody	**GetACLACL**[RF]
(NR)[2]	(NR)[2]	(NR)[2]	**DeleteMethodACL**
(NR)[2]	(NR)[2]	(NR)[2]	**AddACELog**
(NR)[2]	(NR)[2]	(NR)[2]	**RemoveACELog**
(NR)[2]	(NR)[2]	(NR)[2]	**GetACLLog**
(NR)[2]	(NR)[2]	(NR)[2]	**DeleteMethodLog**
(NR)[2]	(NR)[2]	(NR)[2]	**LogTo**

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			Table Association – informative only
(ND)[3]	(ND)[3]	(ND)[3]	**UID**
00 00 08 08 00 00 00 03[7]	00 00 08 08 00 00 00 02[6]	00 00 08 08 00 00 00 02[6]	**InvokingID**[RF]
RBFWDownload[1]	FWDownload[1]	FWDownload[1]	InvokingID Name
Set	Get	Set	**MethodID**
(NR)[2]	(NR)[2]	(NR)[2]	**CommonName**
Note 8	ACE_Anybody	ACE_SID_Set_Port	**ACL**[RF]
(NR)[2]	(NR)[2]	(NR)[2]	**Log**
(NR)[2]	(NR)[2]	(NR)[2]	**AddACEACL**
(NR)[2]	(NR)[2]	(NR)[2]	**RemoveACEACL**
ACE_Anybody	ACE_Anybody	ACE_Anybody	**GetACLACL**[RF]
(NR)[2]	(NR)[2]	(NR)[2]	**DeleteMethodACL**
(NR)[2]	(NR)[2]	(NR)[2]	**AddACELog**
(NR)[2]	(NR)[2]	(NR)[2]	**RemoveACELog**
(NR)[2]	(NR)[2]	(NR)[2]	**GetACLLog**
(NR)[2]	(NR)[2]	(NR)[2]	**DeleteMethodLog**
(NR)[2]	(NR)[2]	(NR)[2]	**LogTo**

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			Table Association – informative only
(ND)[3]	(ND)[3]	(ND)[3]	**UID**
00 00 08 08 00 00 00 04[9]	00 00 08 08 00 00 00 04[9]	00 00 08 08 00 00 00 03[7]	**InvokingID**[RF]
ManufacturerLog[1]	ManufacturerLog[1]	RBFWDownload[1]	InvokingID Name
Get	Set	Get	**MethodID**
(NR)[2]	(NR)[2]	(NR)[2]	**CommonName**
ACE_Anybody	ACE_SID_Set_Port	ACE_Anybody	**ACL**[RF]
(NR)[2]	(NR)[2]	(NR)[2]	**Log**
(NR)[2]	(NR)[2]	(NR)[2]	**AddACEACL**
(NR)[2]	(NR)[2]	(NR)[2]	**RemoveACEACL**
ACE_Anybody	ACE_Anybody	ACE_Anybody	**GetACLACL**[RF]
(NR)[2]	(NR)[2]	(NR)[2]	**DeleteMethodACL**
(NR)[2]	(NR)[2]	(NR)[2]	**AddACELog**
(NR)[2]	(NR)[2]	(NR)[2]	**RemoveACELog**
(NR)[2]	(NR)[2]	(NR)[2]	**GetACLLog**
(NR)[2]	(NR)[2]	(NR)[2]	**DeleteMethodLog**
(NR)[2]	(NR)[2]	(NR)[2]	**LogTo**

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Table Association – informative only	UID	InvokingID[RF]	InvokingID Name	MethodID	CommonName	ACL[RF]	Log	AddACEACL	RemoveACEACL	GetACLACL[RF]	DeleteMethodACL	AddACELog	RemoveACELog	GetACLLog	DeleteMethodLog	LogTo

Notes:

1. The text in the table cell is informative. See the Core Specification [2] and the applicable Opal Family SSC.
2. The content of the cell is defined by the Core Specification [2] and the applicable Opal Family SSC.
3. The content of the cell is not defined.
4. Support of this row is required only if the ACE is required in the `AccessControl` table (see section 5.3.1.4).
5. Support of this row is required only if the Diagnostic Logical_Port object is supported by the TPer (see section 5.3.1.2).
6. Support of this row is required only if the FWDownload Logical_Port object is supported by the TPer (see section 5.3.1.2).
7. Support of this row is required only if the RBFWDownload Logical_Port object is supported by the TPer (see section 5.3.1.2).
8. The value of the Set method ACL column SHALL contain either the ACE_SID_Set_Port, ACE_PSID_Set_Port, or ACE_SID_PSID_Set_Port ACE object (see section 5.3.1.3).
9. Support of this row is required only if the ManufacturerLog Logical_Port object is supported by the TPer (see section 5.3.1.2).
10. The value of the Set method ACL column SHALL contain either the ACE_SID_Set_Port or ACE_SID_Makers_Set_Port ACE object (see section 5.3.1.3).
11. RF: The content of each cell in this column is Read-only as specified as specified by section 5.3.1.4

5.4 Locking SP

This feature set requires no additions or modifications to the Locking SP.

5.5 Additional SPs

This feature set requires no additional SPs.